Calculus I. Partial 1: Limits and the derivative as a limit process

Francisco Vazquez-Tavares

Jan-May 2026. Recent compilation: January 19, 2026

Class conditions

- There are no dumb questions
- Cellphones at the professor's desk at the begginig of class
- If you want to use the cellphone, take it from the desk and go outside
- Bring notebook
- All activities and homeworks are deliver by hand
- Late deliveries: Two days after due date with grade over 80
- No electronic devices
- Web assign is for home
- Most of homeworks are webassign and exam corrections

Introduction to the course

Important rules You need to leave your cellphones at the professor's desk at the beginning of the session. If you leave your cellphone at the professor's desk at the beginning of the session, you will get the attendance. If you really need to use it, take it and go outside the classroom. If you use your cellphone during the first part of the session for more than 4 minutes, you will not receive the attendance. If you are here for the first part of the sessions and then you leave after the break, you still have attendance. However, if it is a phone call from your parents, that might be an exception. I have mandalas if you need to do something with your hands.

Main time distribution Generally the classes will be divided into three sections: lecture, break, and workshop. During the lecture time block, I'll introduce the topics, show procedures, or provide feedback, depending on the day. The break will be 5 minutes, and you can use your cellphone, go to the restroom, or take a quick snack. Finally, in the workshop session, there are going to be 3 activities with due dates before ending class. It doesn't matter if you arrive at the correct answer; the most important thing is that you practice math. The lecture and workshop sessions are planned to be 20 minutes each but might change depending on the topics; meanwhile, the break will be 5 minutes.

Exams will be on Monday. Fridays will be more like a recap session and workshop to resolve questions related to WebAssign activities. Now, there is an announcement in Canvas about how to enroll yourselves into the WebAssign platform; you will have until this Friday, January 16th.

Evaluation system The final grade is composed as follows,

1. Partial 1: 28%
2. Partial 2: 28%

3. Final period exam: 30%
4. Final Period quiz: 4%
5. Final webassign activities: 3%
6. Final period activities: 3%
7. Final period core activity: 4%

Each period will be graded as afollows,

- Partial exam: 50%
- Quizzes: 15%
- Collaborative activities: 15%
- Activities: 10%
- Webassign: 10%

Important dates

- Interpartial exam 1: January 26
- Interpartial exam 2: February 9
- Partial exam 1: February 16
- Interpartial exam 3: March 9
- Partial exam 2: March 23
- Interpartial exam 4: April 27
- Partial exam: May 12

Introduction to Calculus

At some point in the history of humankind there were two open questions: How do you determine the slope of a curve at a single, exact point? And how do you measure the area of a shape with a curved boundary? The first question was asked by the Greeks. They were studying lines that only touch a circle once, most commonly known as the tangent line to a circle. However, for other kinds of graphs it was a mystery, like for a straight line, a parabola, or an ellipse. After some centuries, the answer to this question led us to the development of the derivative. On the other hand, the second question led to the creation of the integral. In a more specific tale, Archimedes was a master of computing areas under parabolas; however, his method was a headache. Some years later, astronomers like Kepler needed to calculate areas of planetary orbits, and this was solved by the definition of an integral.

In general terms, the derivative allows us to give a quantitative answer to questions about the rate of change at an instant, like “How fast is it changing right now”? The integral allows us to give a quantitative answer to questions about the accumulation of change, like “How much has accumulated in total over time”? These two questions are fundamentally linked. If you have a graph of speed over time, the derivative gives the acceleration at any moment, and the integral gives the total distance traveled.

In this course, we only are going to focus on the first question, "*How do you determine the slope of a curve at a single, exact point?*" The answer of this question is going to lead us to the definition of the derivative and that concept it is usefull at GPS navigation, medicine dosing, video games, econocmis, climate models, and other disciplines.

Hence, some of the applications that we are going to explore at the end of the course are related speed, growth, decay, accumulation or in a more abstract concept, flux.

The big picture Calculus is related with arithmetic in the sense that the addition operation finds the total and the substraction operation finds a difference. In calculus, the integration operation finds a total, meanwhile the derivative finds a rate of change. The reason of why calculus is more related with arithmetic is because arithmetic give us tools to quantify **static** amounts, meanwhile calculus give us tools to quantify **dynamic**, changing amounts.

This is quite different from the math that you learn from previous courses. In math 3 and math 4 it was more focused on algebra. And algebra is more like an architect who designs the blueprint, that is the function. Calculus is the surveyor and engineer who uses specialized tool to take precises measurements from that blue print.

For example, algebra give us a function that models the height $h(t) = -5t^2 + 20t + 1$. And can also allow us to answer the question "*When does the ball hit the ground?*" by solving $h(t) = 0$. Calculus is going to help us to answer questions like "*What is the ball's velocity at the exact instant?*" with the derivative and "*What is the total distance traveled by the ball between and interval of time?*" with integration.

As a wrap up, arithmetic allow us to handle numbers. Algebra is to handle unknowns and relationships. And now, calculus will help us to handle change itself. Finally, the main association that I want you to keep in mind is the following: Derivatives answers "*How fast right now*" and the integrals answers *How much in total*. On the other hand, I also want to make an spoiler. The integral and derivative are two sides of the same coin or also known as inverse operations.

Recap

Let's start by recalling the following algebraic properties,

Algebra properties

$$a+b=b+a, \quad (a+b)+c=a+(b+c), \quad a(b+c)=ab+ac,$$
$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}, \quad \frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}, \quad \frac{-a}{b}=-\frac{a}{b}=\frac{a}{-b}, \quad \frac{\frac{a}{b}}{\frac{c}{d}}=\frac{ad}{bc}$$
$$\sqrt{ab}=\sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}}=\frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a}=a^{1/n}$$
$$a^2-b^2=(a-b)(a+b)$$

Laws of exponents

$$a^r\cdot a^s=a^{r+s}, \quad (a^r)^s=a^{rs}$$
$$\frac{a^r}{a^s}=a^{r-s}, \quad (ab)^r=a^rb^r$$
$$\left(\frac{a}{b}\right)^r=\frac{a^r}{b^r}, \quad b\neq 0$$

Laws of Logarithms

$$\log_a(AB)=\log_a(A)+\log_a(B)$$
$$\log_a\left(\frac{A}{B}\right)=\log_a(A)-\log_a(B)$$
$$\log_a\left(A^C\right)=C\log_a(A)$$

Inverse functions property

$$\left(f\circ f^{-1}\right)(x)=x=\left(f^{-1}\circ f\right)(x) \quad \rightarrow \quad \log_a(a^x)=x=a^{\log_a(x)}$$

Notation

Interval notation. To denote intervals we can use the following notation: $x\in(a,b)$, $x\in[a,b]$ or $a<x<b$. $a\leq x\leq b$. Remember that $(,)$ and $<$ are used for open boundaries, that is that the variable x can not take that value. Meanwhile, $[,]$ and $\leq$ are used for closed boundaries, that is that the variable x can take that value.

Class Activity

$$\text{Simplify}: \frac{x^2}{x-4}-\frac{x+1}{x+2}$$
$$\text{Factor}: x^3y-4xy$$
$$\text{Solve}: \frac{2x}{x+7}=\frac{2x-7}{x}$$
Function compostion $(f\circ g)(x)$;
$f(x)=x^2+3x-8$; $g(x)=3x-5$

Interpretation of math

Let's assume that we can quantify the relation between money that is needed to print a book with a function $f(x)$. And we can quantify the relation between the money that we earn by selling a book $g(x)$. We can create an equation that allows us to find the money that is equal to selling and printing books.

Now, let's imagine that the following algebraic expression represents the situation above,

$$\frac{2x}{x+7} = \frac{2x-7}{x}.$$

Imagine that the term $\frac{2x}{x+7}$ quantifies the cost of printing a book and the term $\frac{2x-7}{x}$ quantifies the amount of money that is earned by selling books. We are going to find the value of x that satisfies this condition.

In order to find this condition, we can manipulate the expression using algebraic properties to the following,

$$7x - 49 = 0.$$

We have translated the problem of finding the intersection of two lines or functions into finding the roots of a polynomial. This "problem" conversion is very useful, because we know how to find roots of equations with algebraic methods. Which gives us that $x = 7$.

This condition, when we equate one function to another $f(x) = g(x)$, is normally referred to as the "*equilibrium condition*".

Intersections of a function.

Since a good strategy to find equilibrium conditions is to translate the equation into finding roots of polynomials, let's recall what a root of a polynomial is. A root of a polynomial is the value or set of values of the domain that makes the function equal to zero. That is, all the values of x that satisfy the condition $f(x) = 0$.

It is important to remember that a polynomial of degree n has n roots, assuming that the domain of the function, that is x, belongs to the set of complex numbers. However, if the domain is part of the set of real numbers, this is not true for all polynomials. For example, the function $f(x) = x^2 + 0.1$ does not have real roots. This indicates that not every equation with the expression $f(x) = g(x)$ can have a solution that belongs to the set of real numbers. And for an exam exercise, that is a valid and accurate response.

Class Activity

Solve the following equations

$$\frac{x^2+5x+4}{x^2-3x-4} = 0$$
$$x^3 - 3x^2 - 16x + 10 = 3x^2 + 11x + 10$$
$$3^{3x} + 3^x = 3^{2x+1}$$

Change of variable technique

Class of January 15

In this session we will see the change of variable technique to solve an equation. Let's recall the equation,

$$3^{3x} + 3^x = 3^{2x+1}.$$

We can push the idea of "*factoring*" an expression, in the sense that we can algebraically modify the expression to look for similar terms. With this in mind, let's apply the exponential properties to rewrite the equation,

$$(3^x)^3 + 3^x = (3^x)^2 \cdot 3^1.$$

We can see that in all of the terms appears the factor 3^x; hence, we can introduce a "*change of variable*" as follows: $y = 3^x$,

$$\begin{aligned} y^3 + y &= y^2 \cdot 3 \\ y^3 + y - 3y^2 &= 0 \end{aligned}$$

Now, instead of solving an exponential equation, we change the problem into finding the roots of a cubic equation.

$$y\big(y^2 - 3y^1 + 1\big) = 0.$$

Hence, one solution is $y = 0$. The other two solutions can be computed using the general formula, considering that $a = 1,\ b = -3,\ c = 1$,

$$\begin{aligned} y &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(1)}}{2(1)(1)} \\ &= \frac{3 \pm \sqrt{5}}{2}. \end{aligned}$$

Now that we found the three roots of the cubic polynomial, $y \in \{0, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\}$, let's finish the equation by reverting the change of variable. For the first solution we have,

$$\begin{aligned} 3^x &= y = 0 \\ \log_3(3^x) &= \log 3(0) \\ x &= \log 3(0). \end{aligned}$$

which does not exits. Therefore, we get only two solutions left,

$$\begin{aligned} 3^x = y &= \frac{3 \pm \sqrt{5}}{2} \\ \log_3(3^x) &= \log_3\left(\frac{3 \pm \sqrt{5}}{2}\right) \\ x &= \log_3\left(\frac{3 \pm \sqrt{5}}{2}\right). \end{aligned}$$

And those are the intersection points of the original equation.

Class Activities Solve the following equations

$$\begin{aligned} 3^{2x} + 3^x &= 3^{2x+2} \\ e^{2x} - 3e^x + 2 &= 0 \end{aligned}$$

Types of functions. Part 1

In this session we will revisit the types of functions and their main characteristics, such as domain, range, asymptotes, and holes[1]. Furthermore, we will also review the translation of functions. This is becuase by knowing the domain and range of these types of fuctions is going to help in identifying the domain and range of the derivatives.

[1] January 16

Polynomials For polynomials of the form[2]:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0$$

where n is a nonnegative integer.

Polynomial Type	**Domain**	**Range**
Odd degree (n odd)	$(-\infty, \infty)$	$(-\infty, \infty)$
Even degree (n even)	$(-\infty, \infty)$	Depends on leading coefficient:
$a_n > 0$		$[k, \infty)$
$a_n < 0$		$(-\infty, k]$

[2] **Notes:**

- Domain of *all* polynomials is all real numbers, $\mathbb{R} = (-\infty, \infty)$.
- For odd-degree polynomials, the range is always all real numbers due to the end behavior: as $x \to \pm\infty$, $f(x) \to \pm\infty$ (or $\mp\infty$ if $a_n < 0$).
- For even-degree polynomials:
  - If $a_n > 0$, the graph opens upward with a **minimum value** k.
  - If $a_n < 0$, the graph opens downward with a **maximum value** k.
  - k is found using calculus techniques (derivative = 0) or vertex formulas for quadratics.

Exponential Functions For functions of the form[3]:

$$f(x) = a^x \quad \text{or} \quad f(x) = b \cdot a^{kx} + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

Function Type	**Domain**	**Range**
Basic: $f(x) = a^x$	$(-\infty, \infty)$	$(0, \infty)$
With vertical shift: $f(x) = a^x + c$	$(-\infty, \infty)$	(c, ∞)
General: $f(x) = b \cdot a^{kx} + c$	$(-\infty, \infty)$	If $b > 0$: (c, ∞) If $b < 0$: $(-\infty, c)$

[3] **Special Cases**

- **Natural exponential**: $f(x) = e^x$ has domain $\mathbb{R}$ and range $(0, \infty)$.
- **Natural logarithm**: $f(x) = \ln(x)$ has domain $(0, \infty)$ and range $\mathbb{R}$.
- **Exponential with base** $0 < a < 1$: Domain is still $\mathbb{R}$, range is $(0, \infty)$ if no vertical shift.
- **Key relationship**: $f(x) = a^x$ and $g(x) = \log_a(x)$ are inverse functions:

$$\text{Domain of } f = \text{Range of } g = \mathbb{R}$$

$$\text{Range of } f = \text{Domain of } g = (0, \infty)$$

Calculus Connection

- For $f(x) = a^x$, the derivative is $f'(x) = a^x \ln(a)$
- For $f(x) = \ln|x|$, the derivative is $f'(x) = \frac{1}{x}$ (for $x \neq 0$)
- Domain restrictions are crucial when finding derivatives: $\frac{d}{dx}[\ln(u(x))] = \frac{u'(x)}{u(x)}$ only when $u(x) > 0$

Types of functions. Part 2

Logarithmic Functions [4] For functions of the form:

$$f(x) = \log_a(x) \quad \text{or} \quad f(x) = b \cdot \log_a(kx + d) + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

[4] January 19

Function Type	Domain	Range
Basic: $f(x) = \log_a(x)$	$(0, \infty)$	$(-\infty, \infty)$
With horizontal shift: $f(x) = \log_a(x - h)$	(h, ∞)	$(-\infty, \infty)$
General: $f(x) = b \cdot \log_a(kx + d) + c$	Solve: $kx + d > 0$ Typically: $\left(-\frac{d}{k}, \infty\right)$	$(-\infty, \infty)$

Rational functions For rational functions of the form[5]:

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials, $Q(x) \neq 0$. A basic example: $f(x) = \frac{1}{x}$

Function	Domain	Range
$f(x) = \dfrac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$

- **Vertical asymptote** at $x = 0$
- **Horizontal asymptote** at $y = 0$
- **Behavior**: As $x \to 0^-$, $f(x) \to -\infty$; as $x \to 0^+$, $f(x) \to \infty$

  Here are some common patterns and examples

Example	Domain	Range
$f(x) = \dfrac{1}{x^2}$	$(-\infty, 0) \cup (0, \infty)$	$(0, \infty)$
$f(x) = \dfrac{x}{x^2 + 1}$	$(-\infty, \infty)$	$\left[-\frac{1}{2}, \frac{1}{2}\right]$
$f(x) = \dfrac{x^2 - 1}{x^2 - 4}$	$x \neq \pm 2$	$\left(-\infty, \frac{1}{4}\right] \cup (1, \infty)$
$f(x) = \dfrac{x - 1}{x^2 - 1}$	$x \neq \pm 1$ (hole at $x = 1$)	$(-\infty, 0) \cup (0, \infty)$

[5] **Finding Domain and Range: Step-by-Step**

1. **Domain**:
   (a) Factor numerator and denominator completely
   (b) Set denominator $Q(x) = 0$
   (c) Domain: All real numbers except solutions to $Q(x) = 0$
   (d) If factors cancel (holes), note them separately
2. **Range** (requires calculus for certainty):
   (a) Find horizontal asymptotes: Compare degrees of $P(x)$ and $Q(x)$
   (b) Find vertical asymptotes: Zeros of denominator (not cancelled)
   (c) Find critical points: Solve $f'(x) = 0$ or undefined
   (d) Analyze behavior at asymptotes and critical points
   (e) Consider end behavior as $x \to \pm\infty$

Class Activity

Find the domain, range, asymptotes and interceptions of the functions

$$f(x) = \frac{3x^2 - 1}{x^2 - 1}$$

$$g(x) = \log\left[2e^{x-4} - 2\right]$$

$$h(x) = e^{(x^2-2)/x}$$

General Rational Functions

Function Type	Domain	Range
Proper Rational	Exclude all x where	Typically all real numbers
(degree of numerator < degree of denominator)	$Q(x) = 0$ (vertical asymptotes)	except horizontal asymptote value,
		but check with calculus
Improper Rational	Exclude all x where	Typically all real numbers
(degree of numerator $\leq$ degree of denominator)	$Q(x) = 0$	(may have horizontal or oblique asymptote)
With Common Factors	Exclude all x where	Typically all real numbers
(hole in the graph)	$Q(x) = 0$, including holes	except possibly the y-value
		at the hole location
General Case	$\{x \in \mathbb{R} \mid Q(x) \neq 0\}$	Find by:
$f(x) = \dfrac{P(x)}{Q(x)}$		1. Solving $y = f(x)$ for x
		2. Using calculus (derivatives)
		3. Analyzing asymptotes and behavior

Calculus Connections for Range Determination

- Use **first derivative test** to find local extrema
- Use **limits at asymptotes** to determine unbounded behavior

$$\lim_{x \to a^-} f(x) \quad \text{and} \quad \lim_{x \to a^+} f(x) \quad \text{for vertical asymptotes}$$

- Consider **horizontal asymptotes**:
  - If degree of numerator < degree of denominator: $y = 0$
  - If degrees equal: $y = \frac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$
  - If degree of numerator > degree of denominator: No horizontal asymptote (may have oblique asymptote)
- **Holes** occur when factors cancel: The y-value at the hole is found by evaluating the simplified function at that x-value

Important Notes for Calculus I

- Rational functions are continuous on their domains
- Domain restrictions affect where derivatives exist
- When integrating rational functions, consider partial fractions for complex denominators
- Always check domain before applying the Intermediate Value Theorem
- For optimization problems, ensure solutions are in the domain

Algebra with functions

In this final review session, we cover the algebra of functions: how to compute the composition, subtraction, addition, and the inverse of functions[6].

1. Addition and Subtraction of Functions

For two functions f and g, with domains D_f and D_g:

$$(f+g)(x) = f(x)+g(x), \quad (f-g)(x) = f(x)-g(x)$$

Domain: The domain of $f+g$ or $f-g$ is $D_f \cap D_g$.

2. Composition of Functions

The composition $f \circ g$ is defined by:

$$(f \circ g)(x) = f(g(x))$$

Procedure:

1. Compute $g(x)$.
2. Apply f to the result.

Domain: The domain of $f \circ g$ consists of all x in D_g such that $g(x)$ is in D_f.

3. Inverse Functions

A function f has an inverse f^{-1} if it is **one-to-one** (passes the horizontal line test).

Finding f^{-1}:

1. Write $y = f(x)$.
2. Interchange x and y.
3. Solve for y.
4. Write $f^{-1}(x) = y$.

Properties:

$$f(f^{-1}(x)) = x \quad \text{and} \quad f^{-1}(f(x)) = x$$

Domain and Range: The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f.

[6] January 20

Class Activity

Compute the following operations with functions considering that $f(x) = e^x, g(x) = -(x-1)^2+1$

$$h(x) = (f \circ g)(x)$$
$$j(x) = (g \circ f)(x)$$

Compute: $h^{-1}(x)$

Compute: $\dfrac{(g \circ j)(x)}{2}$

Solutions of previous class activity:

$$f(x) = \frac{3x^2-1}{x^2-1}.$$

Asymptotes: $y = 3; \quad x = \pm 1$

Domain: $x \in (-\infty,-1) \cup (-1,1) \cup (1,\infty)$

Range: $f(x) \in (-\infty,3) \cup (3,\infty)$

y inteceptions: $f(0) = 1$

x inteceptions: $x = \pm \dfrac{1}{\sqrt{3}}$

$$g(x) = \log\left[2e^{x-4}-2\right].$$

Asymptotes: $x = 4$

Domain: $x \in (4,\infty)$

Range: $f(x) \in (-\infty,\infty)$

y inteceptions: No exist

x inteceptions: $x = \ln\left[\dfrac{3}{2}+4\right]$

$$h(x) = e^{(x^2-2)/x}.$$

Asymptotes: $y = 0$

Domain: $x \in (-\infty,\infty)$

Range: $f(x) \in (0,\infty)$

y inteceptions: No exist

x inteceptions: No exist

Limits. Part 1

Limits describe how functions behave near specific points. The concept of "*Limit*" it is invented as a tool to understand trends, holes and instantaneous change. This allow to,

- Define instantaneous rates of change (Derivatives)
- Calculate accumulated change (integrals)
- Analyze function behaviour near discontinuities or asymptotes

January 21

Solutions of previous class activity:

$$f(x) = e^x, \quad g(x) = -(x-1)^2 + 1$$

$$\begin{aligned} h(x) &= e^{-(x-1)^2+1} \\ j(x) &= -(e^x - 1)^2 + 1 \\ h^{-1}(x) &= \pm\sqrt{-\ln[x]+1} + 1 \\ \frac{(g \circ j)(x)}{2} &= \frac{-(e^x-1)^4+1}{2} \end{aligned}$$

The derivative can be think of as a "*Instantaneous velocity*", which is the limit of your average velocity as the time interval shrinks to zero. Without limits, we can only talk about average speed over an interval or approximate areas.

For example, let's assume that a ball is dropped from rest and its distance fallen is $d(t) = 4.9t^2$ meters. Let's try to compute the speed without limits for the time $t = 3$.

$$s(t) = \frac{d(t + \Delta t) - d(t)}{\Delta t},$$

where Δ is the time interval that we are considering. The average speed is only an estimate. The ratio becomes closer to $29.4\,\mathrm{m\,s^{-1}}$ as the time between them gets shorter[7] , but we can never use this method to find the exact speed at t=3. Why? The formula (change in d) / (change in t) needs two points and an interval. At the single moment $t = 3$, there is no change in t, and dividing by zero is not possible. We can only make better and better guesses, but we can't say what the exact amount is that they are getting closer to.

[7]

Interval	Avg. Speed	Result
$[3,4]$	$\frac{d(4)-d(3)}{1}$	34.3 m/s
$[3,3.5]$	$\frac{d(3.5)-d(3)}{0.5}$	31.85 m/s
$[3,3.1]$	$\frac{d(3.1)-d(3)}{0.1}$	29.89 m/s
$[3,3.01]$	$\frac{\Delta d}{0.01}$	29.449 m/s
$[3,3.001]$	$\frac{\Delta d}{0.001}$	29.4049 m/s

$d(t) = 4.9t^2$ m
$d(3) = 44.1$ m
$d(4) = 78.4$ m
$d(3.5) = 60.025$ m
$d(3.1) = 47.089$ m

On the other hand, by using limits, we can get the exact value of the instantaneous velocity, as follows, Starting from the previous average speed equation, however replacing the arguments of the distance function by $3 + h$ and 3, we get,

$$s(t) = \frac{d(3+h) - d(3)}{h}.$$

h now represents the time interval, and we are going to take the limit as this time interval goes to 0. Next, we evaluate the distance function at the given arguments[8],

[8]

$$\begin{aligned} s(t) &= \frac{4.9(3+h)^2 - 4.9(3)^2}{h} \\ &= \frac{4.9(9+6h+h^2) - 4.9(9)}{h} \\ &= \frac{44.1 + 29.4h + 4.9h^2 - 44.1}{h} \\ &= \frac{29.4h + 4.9h^2}{h}. \end{aligned}$$

$$s(t) = \frac{29.4h + 4.9h^2}{h}.$$

Since $h \neq 0$, because we are going to take the limit approaching 0,

$$s(t) = 29.4 + 4.9h.$$

Now, we compute the limit as h tends to zero,

$$\begin{aligned} \lim_{h\to 0} s(t) &= \lim_{h\to 0} [29.4 + 4.9h] \\ &= 29.4. \end{aligned}$$

Hence, the instantaneous velocity at $t = 3$ is $29.4\,\mathrm{m\,s^{-1}}$.

Limits. Part 2

January 22

In this session we are going to learn how to compute limits with a numerical approach and understand the main difference of computing the limit and just evaluate a function.

By evaluating a function, it answers to the question "*What is the function's value at this point?*" and it just inserting a number directly to the algebraic expression. On the other hand, taking a Limit allows to answer to "*What value does the function approach as we get arbitrarily close to a point?*" And the function doesn't even have to be defined at the point itself.

For example, let's consider

$$f(x) = \frac{x^2-1}{x-1}.$$

If we evaluate the function at $x = 1$ we get $0/0$, which is undefined. Furthermore, since is $0/0$, this means that at $x = 1$ the function has a hole.

In order to known the value that the function is approaching when the domain goes to 1, let's compute the limit using a numerical method. First lets build a table with values that are near 1, for instance,

x	$f(x) = \frac{x^2-1}{x-1}$
1.5	$\frac{(1.5)^2-1}{1.5-1} = \frac{2.25-1}{0.5} = \frac{1.25}{0.5} = 2.5$
1.3	$\frac{(1.3)^2-1}{1.3-1} = \frac{1.69-1}{0.3} = \frac{0.69}{0.3} = 2.3$
1.1	$\frac{(1.1)^2-1}{1.1-1} = \frac{1.21-1}{0.1} = \frac{0.21}{0.1} = 2.1$
1.01	$\frac{(1.01)^2-1}{1.01-1} = \frac{1.0201-1}{0.01} = \frac{0.0201}{0.01} = 2.01$
1.001	$\frac{(1.001)^2-1}{1.001-1} = \frac{1.002001-1}{0.001} = \frac{0.002001}{0.001} = 2.001$
1.000001	$\frac{(1.000001)^2-1}{1.000001-1} = \frac{1.000002000001-1}{0.000001} = \frac{0.000002000001}{0.000001} = 2.000001$

With this we can say that the "*limit*" tells us the value the function is trying to be at $x = 1$, even though it never actually reaches it there.

Class Activity

Compute the limit with the numerical approach of the following functions,

$$f(x) = \frac{1}{x} \quad x \to 0$$
$$g(x) = \frac{x^2-4}{x-2} \quad x \to 2$$
$$h(x) = \frac{\sin(x)}{x} \quad x \to 0$$

Limits. Part 3

January 23

In this session we will see the graphical approach of a limit.

Piecewise functions and stuff

Typical exercises

- Expand and simplify
- Complete the squared or Factor the expression
- Evaluate without using calculator
- Solve the equation
- Composition of functions
- Evaluate a function
- Domain and range
- Find a linear equation given two points
- Translation of a function

Intercepts of a function

References

James Stewart, Dan Clegg, and Saleem Watson. "*Calculus early transcendentals James Stewart, Daniel Clegg, Saleem Watson*". Cengage, 2024.